

SHOPFORGE

Ecommerce Platform

Software Requirements Specification (SRS)

Version 1.0 | Stack: Spring Boot + PostgreSQL + React

Document Status: APPROVED FOR DEVELOPMENT

1. Introduction

1.1 Purpose

This SRS defines functional and non-functional requirements for ShopForge — a full-featured B2C ecommerce platform built on Spring Boot (backend REST API), PostgreSQL (primary database), and React (frontend SPA). It serves as contract between stakeholders and development team.

1.2 Scope

ShopForge covers:

- Customer-facing storefront with product catalog, cart, checkout, order tracking
- Seller/Admin portal for product, order, inventory management
- Authentication (JWT + OAuth2), role-based access control
- Payment gateway integration (Stripe/Razorpay)
- Email notifications, search with filters, reviews and ratings
- RESTful API consumable by mobile clients

1.3 Definitions

Term	Definition
SKU	Stock Keeping Unit — unique product variant identifier
JWT	JSON Web Token — stateless auth token
OTP	One-Time Password for phone/email verification
PDP	Product Detail Page
PLP	Product Listing Page / Category page
Cart	Session or DB-persisted list of items before checkout
Order	Confirmed purchase record with payment and fulfillment status
Wishlist	User-saved products for future purchase
Admin	Platform superuser — full access
Seller	Merchant who lists products on platform

1.4 Tech Stack

Layer	Technology	Version
Backend API	Spring Boot	3.2.x
Language	Java	21 (LTS)
Database	PostgreSQL	16.x
ORM	Spring Data JPA / Hibernate	6.x
Auth	Spring Security + JWT + OAuth2	—
Cache	Redis	7.x
Search	Elasticsearch (optional) / PostgreSQL FTS	8.x
File Storage	AWS S3 / MinIO	—
Messaging	RabbitMQ / Spring Events	3.x
Frontend	React + TypeScript + Tailwind	18.x

Layer	Technology	Version
API Docs	SpringDoc OpenAPI (Swagger)	2.x
Containerization	Docker + Docker Compose	—
CI/CD	GitHub Actions	—

2. System Overview

2.1 Architecture

Three-tier architecture: React SPA (Vercel/Nginx) → Spring Boot REST API (Docker/EC2) → PostgreSQL + Redis (RDS/ElastiCache). Stateless API layer enables horizontal scaling. JWT stored in HttpOnly cookies. Media assets served via CDN from S3.

2.2 User Roles

Role	Description	Key Permissions
Guest	Unauthenticated visitor	Browse, search, view products, register
Customer	Registered buyer	Cart, checkout, orders, wishlist, reviews
Seller	Merchant account	Manage own products, view own orders, inventory
Admin	Platform operator	Full CRUD on all entities, user management, reports
Support	CS agent	View orders, process refunds, manage tickets

3. Functional Requirements

3.1 Authentication & Authorization

3.1.1 Registration

- Email + password signup with email verification OTP
- Google OAuth2 SSO
- Phone number optional — OTP via SMS for checkout
- Password: min 8 chars, 1 uppercase, 1 digit, 1 special

3.1.2 Login

- Email/password login → JWT access token (15 min) + refresh token (7 days)
- Refresh token rotation — invalidate on use
- Remember Me — 30-day refresh token
- Max 5 failed attempts → 15-min logout

3.1.3 Password Management

- Forgot password → email reset link (expire 1 hr)
- Change password requires current password
- Force re-login all sessions on password change

3.2 Product Catalog

3.2.1 Product Management (Admin/Seller)

- Create product: name, description (rich text), brand, category, tags
- Multiple images (up to 8) — primary image required
- Product variants: size, color, material — each variant has own SKU, price, stock
- Base price, sale price, tax class
- SEO fields: meta title, meta description, slug (auto-generated, editable)
- Status: DRAFT, ACTIVE, INACTIVE, ARCHIVED
- Bulk import via CSV/Excel

3.2.2 Product Listing (Customer/Guest)

- PLP with pagination (default 24/page), infinite scroll option
- Filter by: category, brand, price range, rating, availability, attributes
- Sort by: relevance, price asc/desc, newest, best selling, rating
- PDP: images gallery, variant selector, stock status, estimated delivery
- Recently viewed (localStorage + DB for logged-in users)

3.3 Search

- Full-text search across name, description, brand, category, tags
- Autocomplete suggestions (debounced 300ms)
- Search history per user (last 10)
- No-results page with suggestions
- PostgreSQL tsvector FTS (Phase 1) — Elasticsearch migration optional (Phase 2)

3.4 Cart & Wishlist

3.4.1 Cart

- Guest cart (sessionStorage) merged on login
- Persistent cart (DB) for logged-in users
- Add/update quantity/remove items
- Stock validation on add and at checkout
- Apply coupon codes
- Cart expiry: 30 days inactive

3.4.2 Wishlist

- Multiple wishlists per user (default: 1)
- Move to cart from wishlist
- Share wishlist via link (public/private toggle)

3.5 Checkout

1. Address selection or new address entry
 2. Shipping method selection (rates from config)
 3. Coupon/promo code application
 4. Order summary review
 5. Payment
 6. Order confirmation + email
- Save multiple addresses per user
 - Address validation (PIN code → city/state autofill)
 - Guest checkout allowed (email required)

3.6 Payment

Gateway	Methods	Notes
Razorpay	UPI, Cards, Net Banking, Wallets	Primary for INR
Stripe	Cards, Apple Pay, Google Pay	International
COD	Cash on Delivery	Configurable by PIN code / order value

- Webhook processing for async payment confirmation
- Payment retry for failed transactions

- Partial payment / installments (Phase 2)

3.7 Order Management

3.7.1 Customer

- View order list with status filter
- Order detail: items, price breakdown, timeline
- Cancel order (before SHIPPED)
- Return/exchange request (within return window)
- Track shipment with tracking number link
- Download invoice (PDF)

3.7.2 Admin/Seller

- Order list with bulk status update
- Assign fulfillment / print packing slip
- Update tracking number
- Process refunds (full/partial)

3.7.3 Order Status Flow

Status	Description	Next States
PENDING	Payment initiated	PAYMENT_RECEIVED, PAYMENT_FAILED, CANCELLED
PAYMENT_RECEIVED	Payment confirmed	PROCESSING, CANCELLED
PROCESSING	Seller preparing order	SHIPPED, CANCELLED
SHIPPED	Handed to courier	DELIVERED, RETURN_REQUESTED
DELIVERED	Customer received	RETURN_REQUESTED, COMPLETED
COMPLETED	Return window expired	—
CANCELLED	Order cancelled	REFUND_INITIATED
RETURN_REQUESTED	Return initiated	RETURN_APPROVED, RETURN_REJECTED
REFUND_INITIATED	Refund processing	REFUNDED
REFUNDED	Money returned	—

3.8 Reviews & Ratings

- Rating 1-5 stars + text review (min 20 chars)
- One review per product per verified purchase
- Image upload with review (up to 3)
- Helpful votes on reviews
- Admin moderation (approve/reject/flag)
- Aggregate rating computed with Bayesian average

3.9 Notifications

Trigger	Channel	Recipient
Order placed	Email + SMS	Customer
Payment confirmed	Email	Customer
Order shipped	Email + SMS + Push	Customer
Order delivered	Email	Customer
Return approved	Email	Customer
Refund processed	Email	Customer
Low stock	Email	Seller/Admin
New order	Email	Seller

3.10 Admin Dashboard

- KPI widgets: GMV, orders today, revenue, new users
- Sales charts: daily/weekly/monthly revenue
- Top selling products, top categories
- User management: list, view, ban, roles
- Category management: tree structure, add/edit/delete
- Coupon/promotion management
- Inventory alerts (low stock threshold config)
- Report export: orders, revenue, inventory (CSV/Excel)

3.11 Coupon & Promotions

Type	Description
PERCENTAGE	X% off total or specific categories/products
FIXED_AMOUNT	Flat Rs./\$ off

Type	Description
FREE_SHIPPING	Waive shipping charges
BUY_X_GET_Y	Buy X quantity get Y free
FLASH_SALE	Time-bounded product price overrides

- Coupon: code, type, value, min order value, max discount cap, usage limit, expiry
- One coupon per order
- Per-user usage limit

4. Non-Functional Requirements

Category	Requirement
Performance	API p95 < 300ms under 1000 concurrent users
Availability	99.9% uptime SLA (excluding planned maintenance)
Scalability	Horizontal pod scaling; DB read replicas for reporting
Security	OWASP Top 10 compliance; PCI-DSS for payment data
Data Integrity	DB transactions for all financial ops; idempotency keys on payments
SEO	Server-side rendering or pre-rendering for product pages
Accessibility	WCAG 2.1 Level AA on frontend
Compliance	GDPR-ready: data export, deletion requests
Audit	All admin actions logged with actor + timestamp + diff
Backup	Daily DB snapshots; PITR enabled; 30-day retention

5. API Design Guidelines

- RESTful, versioned: /api/v1/...
- JSON request/response; Content-Type: application/json
- Auth: Bearer JWT in Authorization header OR HttpOnly cookie

- Pagination: ?page=0&size=24&sort=createdAt,desc (Spring Pageable convention)
- Error envelope: { status, code, message, errors[] }
- Idempotency-Key header for payment and order creation endpoints
- Rate limiting: 100 req/min guest, 300 req/min authenticated

5.1 Core Endpoints Summary

Module	Base Path	Key Endpoints
Auth	/api/v1/auth	POST /register, /login, /refresh, /logout, /forgot-password
Users	/api/v1/users	GET /me, PATCH /me, GET /{id} (admin)
Products	/api/v1/products	GET /, GET /{id}, POST / (seller), PATCH /{id}, DELETE /{id}
Categories	/api/v1/categories	GET /tree, POST /, PATCH /{id}
Cart	/api/v1/cart	GET /, POST /items, PATCH /items/{id}, DELETE /items/{id}
Orders	/api/v1/orders	POST /, GET /, GET /{id}, PATCH /{id}/status
Payments	/api/v1/payments	POST /initiate, POST /webhook, GET /{orderId}
Reviews	/api/v1/reviews	POST /, GET /product/{id}, PATCH /{id}, DELETE /{id}
Addresses	/api/v1/addresses	GET /, POST /, PATCH /{id}, DELETE /{id}
Wishlist	/api/v1/wishlists	GET /, POST /items, DELETE /items/{productId}
Coupons	/api/v1/coupons	POST /validate, POST / (admin), GET / (admin)
Admin	/api/v1/admin	GET /dashboard, GET /reports/*, POST /users/{id}/ban

6. Constraints & Assumptions

- Initial launch targets India market (INR, Razorpay primary)
- Single-warehouse inventory model for Phase 1

- No multi-currency for Phase 1
- Mobile apps (Flutter) consume same API — no API changes break mobile
- All dates stored UTC, displayed in user timezone
- Images compressed to WebP on upload via background job

7. Development Milestones

Sprint	Duration	Deliverables
Sprint 1	Week 1-2	Project setup, DB schema, auth (register/login/JWT), user API
Sprint 2	Week 3-4	Product CRUD, category tree, image upload to S3, search (FTS)
Sprint 3	Week 5-6	Cart, wishlist, address management, shipping config
Sprint 4	Week 7-8	Checkout flow, payment integration (Razorpay), order creation
Sprint 5	Week 9-10	Order management, status flow, notifications (email/SMS)
Sprint 6	Week 11-12	Reviews, coupons, admin dashboard, reports
Sprint 7	Week 13-14	Performance tuning, Redis caching, Elasticsearch (optional)
Sprint 8	Week 15-16	Security audit, test coverage, Docker packaging, staging deploy